Yinda Chen

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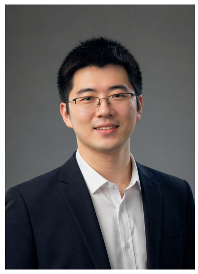
https://ydchen0806.github.io/

https://scholar.google.com/citations?user=hCvlj5cAAAAJ&hl=en&oi=ao



Education

University of Science and Technology of China & Shanghai AI Lab — Hefei & Shanghai

Information and Communication Engineering, Ph.D. Candidate — **2024.7 ~ 2027.6 (Expected)**

- **Research Focus:**Unified generative-understanding models, world models, brain-inspired intelligence
- **Advisors:**Prof. Feng Wu (CAE Academician, IEEE Fellow), Prof. Zhiwei Xiong; Co-supervisor: Prof. Xiaoou Tang (IEEE Fellow, Shanghai AI Lab)
- **Core Courses:**Algorithm Design and Analysis, Statistical Learning, Deep Learning, Reinforcement Learning
- **Honors:**Principal Investigator of National Natural Science Foundation Ph.D. Project (2024)

University of Science and Technology of China — Hefei

Computer Technology, M.S. — **2022.9 ~ 2024.7**

- **Honors:**National Graduate Scholarship (2022)

Xiamen University — Xiamen

Environmental Ecological Engineering & Economics (Dual Degree), B.S. — **2018.9 ~ 2022.7**

- **Overall Ranking: 1/31**
- **Honors:**Xiamen University Academic Star (2021), CDA Level 1 Certification (2022), Kaggle Expert
- **Advisor:**Prof. Yuanye Zhang

Research Projects

Pelican-Unified 1.0: Unified Embodied Intelligence Model — 2026.5

Technical Report *Core Contributor (Ranked Second)*

- Contributed to a unified embodied intelligence model for **understanding, reasoning, imagination, and action**.
- Helped build a shared framework that integrates multimodal understanding, chain-of-thought reasoning, future video rollout, and low-level action generation.
- Paper: arXiv PDF
- Media coverage: People's Daily, Beijing Daily / Beijing Gov, Machine Heart, Zhidongxi

National Natural Science Foundation Ph.D. Project — 2025.1 ~ 2027.12

NSF Project *Principal Investigator (Funding: 300k RMB, Only recipient in Information field, Anhui Province, 2024)*

[1] Foundation **10-billion-scale** brain neural **foundation model** training
- Neuron underlying visual feature **token construction**
- Neural architecture search, **autoregressive visual model** training

[2] **Fine-grained** neuron **segmentation**
- **Skeleton features**, attention decomposition mechanisms
- Complementary masking, segmentation **robustness enhancement**

[3] Neural **circuit reconstruction**
- Point cloud morphological features and image feature **alignment**
- Candidate competition strategy, **progressive reasoning**, long-range tracking

Large-scale Self-supervised Pretraining — 2022.5 ~ 2023.12

Conference&Journal *First Author*

[1] 【LONG ORAL】**Self-supervised neuron segmentation with multi-agent reinforcement learning**, **IJCAI 23** **[CCF-A]**
- Improved MAE masking strategy using reinforcement learning, automatically selecting mask ratios and schemes.
- Introduced multi-agent framework for more efficient self-supervised feature learning, improving segmentation accuracy by 12%.

[2] **MaskTwins: Dual-form Complementary Masking for Domain-Adaptive Image Segmentation**, **ICML 25** **[CCF-A]**
- Proposed new complementary masking theory from sparse signal reconstruction perspective, rigorously proving theoretical advantages of dual-form complementary masking in extracting domain-invariant features.
- Designed simple and efficient unsupervised domain adaptation framework, achieving cross-domain image segmentation through complementary masking consistency learning, improving natural image segmentation by 2.7% mIoU and biomedical image segmentation by 3.2% IoU.

[3] **TokenUnify: Scalable Autoregressive Visual Pre-training with Mixture Token Prediction**, **ICCV 25** **[CCF-A]**
- Proposed image autoregressive pretraining combined with Mamba framework, demonstrating advantages of long sequences and low computational cost.
- Demonstrated good scaling laws with corresponding theoretical proof, showing logarithmic-linear relationship between model performance and parameters.

[4] EMPOWER: Evolutionary Medical Prompt Optimization With Reinforcement Learning, **JBHI** **[SCI-Q1]**
- Proposed the first evolutionary prompt optimization framework for medical domain, combining domain knowledge and reinforcement learning.
- Core techniques: medical attention, multi-dimensional quality assessment, evolutionary search, and semantic verification for clinical guideline consistency.
- Reduced factual errors by 24.7%, improved domain specificity by 19.6%, and achieved 15.3% higher clinician preference in blind testing.

[5] Learning multiscale consistency for self-supervised electron microscopy instance segmentation, **ICASSP 24** **[CCF-B]**
- Achieved high-performance pretraining strategy based on multiscale feature contrastive learning and feature reconstruction.
- Proposed feature consistency loss function, overcoming representation differences caused by scale variations, improving accuracy by 9%.

[6] [Major Revision] Generative Text-Guided 3D Vision-Language Pretraining for Unified Medical Image Segmentation, Submit to PR
- Generated image descriptions using large language models for multimodal image-text contrastive learning pretraining.
- Innovatively introduced generative text guidance mechanism, solving medical image annotation scarcity problem, achieving zero-shot segmentation.

Large-scale Data Generation and Encoding Compression — 2023.8 ~ Present

Conference&Journal *First Author & Co-first Author*

[1] 【Accepted】Learned Image Coding with Generative Reference of Conditional Latents, **TPAMI** **[SCI-Q1]**
- Extension work of AAAI 25 oral paper, further exploring benefits of reference images for image coding.
- Obtained reference images through local dictionary, network retrieval, and image generation, improving compression performance by 15%.
- Proposed conditional latent generation framework, effectively solving performance degradation when reference images are unavailable.

[2] 【ORAL】Conditional Latent Coding with Learnable Synthesized Reference for Deep Image Compression, **AAAI 25** **[CCF-A]**
- Built image similarity dictionary for retrieving similar images to improve entropy model probability estimation.
- Proposed learnable synthetic reference framework, improving PSNR by 0.6dB at the same bit rate with leading compression performance.

[3] MaskFactory: Towards High-quality Synthetic Data Generation For Dichotomous Image Segmentation, **NeurIPS 24** **[CCF-A]**
- Generated corresponding mask-image pairs using ControlNet through rigid and non-rigid deformation editing of masks.
- Core techniques: deformation-driven mask editing, ControlNet-guided image generation, synthetic data quality assessment.
- Synthetic data performance approached real data in downstream segmentation tasks with only 2% performance gap, significantly reducing annotation costs.

[4] [Major Revision] Joint Semantic and Coded Generation for Conditional Latent Coding, Submit to TCSVT
- Proposed joint semantic and coded generation framework, combining text descriptions with minimal coded features (0.01 bpp) to guide controllable rectified flow models in generating high-fidelity reference images.
- Core techniques: ControlNet for DiT architecture, iterative reference alignment (IRA), dynamic feature fusion (DFS), iterative probability refinement (IPR), theoretical robustness proof.
- Improved compression performance by 15.8% BD-rate compared to VVC, successfully unifying image generation and compression, bridging the gap between semantic and coded representations.

[5] BIMCV-R: A Landmark Dataset for 3D CT Text-Image Retrieval, **MICCAI 24** **[CCF-B]**
- Built the first open-source 3D CT image-text pair dataset containing 10,000 high-quality medical image-description pairs.
- Achieved efficient image-text information retrieval and keyword search, improving recall rate by 25% over traditional methods.

Large Model Engineering — 2023.9 ~ Present

Projects *Core Member*

[1] Image coding and intra-frame prediction large models
- Led design of billion-parameter-scale coding architecture, improving compression rate by 30% over traditional coding standards.

[2] Quantitative large models
- Built LLM-driven workflows for quantitative research, supporting financial text understanding and factor mining.

[3] Medical image segmentation and neuron segmentation large models
- Mainly responsible for pretraining in the team, with experience in large-scale cluster pretraining using 64 A40 GPUs.
- Proficient in large model frameworks like DDP and DeepSpeed, achieving efficient training and optimization of 30-billion parameter models.

Industry Internship Experience

Ubiquant — Beijing — 2026.5 ~ Present

Quantitative Research and Large Language Models *Researcher*

- Working on **quantitative research** and **large language model research**.
- Developing **quantitative large models** for factor mining and data-driven decision making.

Beijing Humanoid Robot Innovation Center (UBTECH) **Beijing** **2025.12 ~ Present**

Embodied Intelligence World Model Algorithm Team *Core Member*

- Focusing on **embodied intelligence world models** and **unified models** for humanoid robots.
- Working on unified multimodal decision-making, **real-robot** closed-loop validation, and **Unify foundation model** training.
- Experienced with **100-GPU H200** cluster training for large-scale embodied foundation models.
- Contributed as a core contributor (ranked second) to **Pelican-Unified 1.0**, a technical report on unified embodied intelligence.

Tencent (IEG) **Shanghai** **2025.8 ~ 2025.12**

Qingyun Program *Research Intern*

- Focused on game video scene understanding technology development, building intelligent video content analysis systems.
- Designed game scene automatic recognition and classification algorithms based on multimodal large models, achieving accuracy above 95%.
- Trained TTS models for Honor of Kings Lingbao (based on f5-tts and indextts2).

PLA General Hospital (301 Hospital) **Beijing** **2023.9 ~ 2024.2**

Data Compression Group *Research Intern*

- Collaborated with Academician Qionghai Dai's team on efficient data compression research.
- Designed medical imaging-specific compression algorithms optimized for CT, MRI modalities, improving compression efficiency by 35%.

Academic Internship Experience

Imperial College London **London (Remote)** **2022.11 ~ 2023.8**

Data Science Institute *RA*

- Collaborated with Associate Professor Rossella Arcucci on multimodal pretraining research, submitting one journal paper.
- Developed image-text contrastive learning framework, achieving 93.5% accuracy on medical diagnosis tasks.

Xiamen University WISER Club **Xiamen** **2021.8 ~ 2022.7**

Data Mining Group *Academic Community Member*

- Responsible for designing and discussing data mining courses, teaching clustering and Transformer sections.
- Mentored 20 undergraduate students to complete machine learning projects, organized 2 campus competition activities.

Wang Yanan Institute of Economics, Xiamen University **Xiamen** **2020.8 ~ 2021.12**

Econometrics *Research Assistant*

- Assisted Associate Professor Jiong Zhu in completing territorial economic statistics, conducting visual feature extraction of homestead information.
- Developed satellite image analysis tools for automatic land use change identification with 85% accuracy.

Honors and Awards

- **National Natural Science Foundation Ph.D. Project** **2024.12**
  - Principal Investigator, Only recipient in Information field, Anhui Province
- **Interdisciplinary Contest in Modeling (ICM), Outstanding Winner** **2024.05**
  - Top 0.17% of 10,388 participating teams worldwide
- **National Graduate Scholarship** **2022.12**
  - Award rate 1%
- **Xiamen University Academic Star** **2021.12**
  - Only undergraduate recipient
- **'Jingrun Cup' Mathematics Competition** **2021.09**
  - First Place, Xiamen University
- **'Internet+' Innovation Competition** **2021.08**
  - Gold Award, Fujian Province
- **National College Student Mathematics Competition Finals** **2021.05**
  - National Second Prize
- **'Challenge Cup' Academic Competition** **2021.05**
  - First Prize, Fujian Province
- **National College Student Mathematics Competition** **2020.11**
  - First Place, Fujian Province

Technical Skills

- **Programming**: Python, C, C++, Java, MATLAB, LaTeX
- **Deep Learning**: PyTorch, TensorFlow, DeepSpeed, DDP
- **Languages**: English (TOEFL 110, GRE 328), Chinese (Native)
- **Tools**: Git, Docker, CUDA, HPC

Professional Service

- **Reviewer**: CVPR 2025, ICCV 2025, WACV 2026, NeurIPS 2024, ICML 2025, ICLR 2024, MICCAI 2025, ACM MM 2024, AISTATS 2024, IJCV, TIP